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Work Experience

2022.11-now **(Pre-tenure) Professor**, *University of Science and Technology of China*, Hefei, China.

2022.01-2022.10 **Associate Professor**, *University of Science and Technology of China*, Hefei, China.

2020.04-2022.01 **Applied Scientist**, *Amazon.com*, Seattle.

Education

2015.05-2020.04 **Phd**, *School of Electrical Engineering and Computer Science, Washington State University, Major: Computer science*, Mentor: Prof. Yinghui Wu.

2012.09–2015.02 **Master**, *School of Computer Science and Engineering, Beihang University, Major: Computer science*, Mentor: Prof. Jinpeng Huai.

2008.09–2012.07 **Bachelor**, *School of Computer Science and Engineering, Beihang University, Major: Computer science*.

Research Interests

- Knowledge construction, knowledge enhanced LLM/time-series models/multimodal models.
- Deep learning (e.g., deep neural network models like CNN and RNN) for graph analysis (e.g., ranking, anomaly detection).
- (Distributed) graph data mining, (temporal) graph query models and languages.

Publications

(55) 2026 Chang Fang, Haokai Wang, Wangqiu Zhou, Jun Jia, Qi Song, Xiangyang Li , *"DeShift: Mechanism-Aware Probabilistic Forecasting under Temporal Distribution Shift."*, Proceedings of the 33th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2027.

(54) 2026 Haokai Wang, Chang Fang, Wangqiu Zhou, Jun Jia, Qi Song, Xiangyang Li , *"Query-Structured Diffusion for Irregular Multivariate Time Series Forecasting."*, Proceedings of the 33th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2027.

(53) 2026 Ruixi Chen, Yuhang Zhang, Ling Zheng, Shixiang Luo, Shanhua Bao, Lei Yu, Qi Song, Xiang-Yang Li, *"Harnessing Large Language Models for Complex-free Protein–Ligand Affinity Prediction via Bimodal Retrieval Enhancement."*, Proceedings of the 33th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2027.

(52) 2026 Yiming Xu, Yuhang Zhang, Wangqiu Zhou, and Qi Song, *"Hi-MEC: Hierarchical Medical Diagnosis via Quality-Aware Knowledge Fusion and Multi-Expert Consultation."*, IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**), 2026.

(51) 2026 Junli Liang, Pengfei Zhou, Rui Lv, Wangqiu Zhou, Qi Song, Xiang-Yang Li, *"CROG: Coverage-aware Reasoning Optimization on Graphs for Multi-hop Question Answering."*, The Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026.

(50) 2026 Qi Zhao, Jian Li, Yizhang Jin, Dongqi Liu, Hang Ding, Qi Song, Xiang-Yang Li, *"RoleArena: A Multi-Agent Role-Playing Environment for Long Multi-Turn Dialogues with Autonomous Plot Progression."*, The Conference on Empirical Methods in Natural Language Processing (**EMNLP** Findings), 2026.

(49) 2026 Yong Liu, Yufan Wang, Chang Fang, Wangqiu Zhou, Qi Song, and Xiangyang Li, *"FDER: Frequency-Decoupled Enhanced Retrieval for Long-term Time Series Forecasting."*, The IEEE International Conference on Data Mining (**ICDM**), 2026.

(48) 2026 Yiming Xu, Yixuan Liu, Yuhang Zhang, Ling Zheng, Yihan Wang, Qi Song, *"MKG-CARE: Case-Aware Reasoning with Multimodal Knowledge Graphs for Explainable Medical Image Diagnosis."*, The Conference on Information and Knowledge Management (**CIKM**), 2026.

(47) 2026 Haokai Wang, Chang Fang, Wangqiu Zhou, Yufan Wang, Qi Song, *"Koopman-Guided Diffusion for Probabilistic Time Series Forecasting."*, The Conference on Information and Knowledge Management (**CIKM**), 2026.

(46) 2026 Hualong Xiao, Gang Luo, Wenqian Ma, Ji Wu, Mingqiang Lin, Bo Liu, Qi Song, *"Efficient screening of retired lithium-ion batteries with a lightweight stacking model via time-frequency data fusion."*, Journal of Energy Storage, 2026.

(45) 2026 Yiming Xu, Jihua Kang, Chunsai Du, Qifan Zhang, Wangqiu Zhou, YiTing Wu, Tianqi Li, Qi Song, *"GVR-Coder: A Visual-Feedback Framework for Structured SVG Generation in Complex Document and Meeting Scenarios."*, 34th ACM International Conference on Multimedia (**MM**), 2026.

(44) 2026 Qi Zhao, Hongyu Yang, Qi Song, Xiangyang Li, *"KnowPath: Leveraging Large Language Model Prior Reasoning Paths for Dynamic Self-Reflective Reasoning via Knowledge Graph Subgraph Exploration."*, Expert Systems With Applications (**ESWA**), 2026.

(43) 2026 Yuhang Zhang, Keyan Ding, Peilin CHEN, Han Liu, Can Lin, Ruixi Chen, Shiqi Wang, Qi Song, *"TIGER: Text-Informed Generalized Enzyme-Reaction Retrieval."*, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026.

(42) 2026 Yunlong Liu, Pengfei Zhou, Hao Wang, Zhiqiang Qin, Ji Wu, Wangqiu Zhou, Gang Luo, Qi Song, Xiangyang Li, *"AReLiB-100k: A Real-World 100k-Scale Dataset and Comprehensive Benchmark for Capacity Estimation of Retired Lithium-Ion Batteries."*, Data Mining and Knowledge Discovery (**ECML-PKDD Journal Track**), 2026.

(41) 2026 Ziwen Wang, Yihan Wang, Junli Liang, Rong Lin, Wangqiu Zhou, Pengfei Zhou, Ling Zheng, Hao Liu, Qi Song, *"All-or-Nothing: Towards Hyperedge-Consistent Node Classification."*, 31st International Conference on Database Systems for Advanced Applications (**DASFAA**), 2026.

(40) 2026 Xinyu Wang, Shenyao Jiang, Hao Zhou, Tianjian Yang, Bo Qian, Ying Liu, Qi Song, Yusheng Ji, *"Chunip: Charging-Uninterrupted In-Band Parallel Communication for Magnetic MIMO Wireless Power Transfer System."*, IEEE International Conference on Computer Communications (**INFOCOM**), 2026.

(39) 2026 Zhenhe Li, Can Lin, Ling Zheng, Wen-Da Wei, Junli Liang, Qi Song, *"GraphIF: Enhancing Multi-Turn Instruction Following for Large Language Models with Relation Graph Prompt."*, The 40th Annual AAAI Conference on Artificial Intelligence (**AAAI**), 2026.

(38) 2025 Xiang-Yang Li, Yihang Wang, Junli Liang, Xinyu Wang, Pengfei Zhou, Qi Zhao, Yuhang Zhang, Qi Song, *"Knowledge-enhanced AI models for domain-specific application: a survey. "*, SCIENCE CHINA Information Sciences (**SCIS**), 2025.

(37) 2025 Yuhang Zhang, Yiming Xu, Peilin Chen, Shiqi Wang, Qi Song, Lei Yu, Wei Cai, *"Knowledge-Enhanced Medical Image Classification via Descriptive Priors from Large Language Models. "*, Health Information Science and Systems (**HISC**), 2025.

(36) 2025 Yuhang Zhang, Peilin Chen, Keyan Ding, Han Liu, Shiqi Wang, Qi Song, *"MPSol: A Multimodal Prompt Learning Framework for Protein Solubility Prediction. "*, IEEE Journal of Biomedical and Health Informatics (**JBHI**), 2025.

(35) 2025 Can Lin, Zhengwang Jiang, Ling Zheng, Qi Zhao, Yuhang Zhang, Qi Song, Wangqiu Zhou, *"RJE: A Retrieval-Judgment-Exploration Framework for Efficient Knowledge Graph Question Answering with LLMs. "*, The Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2025.

(34) 2025 Yiming Xu, Qi Song, Yihan Wang, Wangqiu Zhou and Junli Liang, *"LGC-CR: Few-shot Knowledge Graph Completion via Local Global Contrastive Learning and LLM-Guided Refinement."*, The Conference on Information and Knowledge Management (**CIKM**), 2025.

(33) 2025 Ling Zheng, Qi Song, Yihan Wang, Zhitao Wang, Xiang-Yang Li, *"HE-GAD: A Behavior-Enhanced Contrastive Learning Framework for Graph Anomaly Detection."*, Machine Learning (**ECML-PKDD Journal Track**), 2025.

(32) 2025 Yihan Wang, Qi Song, Ling Zheng, Xiang-Yang Li, *"Efficient and Tractable Conflict Detection in Knowledge Graphs via Path Analysis."*, IEEE Transactions on Big Data (**TBD**), 2025.

(31) 2025 Pengfei Zhou, Yunlong Liu, Junli Liang, Qi Song, Xiangyang Li, *"CrossLinear: Plug-and-Play Cross-Correlation Embedding for Time Series Forecasting with Exogenous Variables."*, Proceedings of the 31th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025.

(30) 2025 Xinyu Wang, Shenyao Jiang, Wangqiu Zhou, Hao Zhou, Zhi Liu, Qi Song, Yusheng Ji, *"In-Band Parallel Feedback Communication for Magnetic MIMO Wireless Power Transfer System."*, IEEE Wireless Communications Magazine, 2025.

(29) 2025 Xinyu Wang, Wangqiu Zhou, Hao Zhou, Shenyao Jiang, Zhi Liu, Xiaoyan Wang, Yusheng Ji, Qi Song, *"Relip: Reliable In-Band Parallel Communication for Magnetic MIMO Wireless Power Transfer System."*, IEEE Transactions on Mobile Computing (**TMC**), 2025.

(28) 2025 Qi Zhao, Qi Song, Tian Xie, Haiyue Zhang, Hongyu Yang, Xiangyang Li, *"Improving Pretrained Language Models with Knowledge Enhancement and Filtering Framework."*, Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL** Findings), 2025.

(27) 2025 Pengfei Zhou, Junli Liang, Yunlong Liu, Ji Wu, Qi Song, Xiangyang Li, *"Capacity estimation of retired lithium-ion batteries using random charging segments from massive real-world data."*, Cell Reports Physical Science (**CRPS**), 2025.

(26) 2025 Xinyu Wang, Wangqiu Zhou and Hao Zhou, Tianjian Yang, Shenyao Jiang, Zhi Liu, Yusheng Ji, Qi Song, *"Fast and Anti-starvation Charging Device Grouping for Magnetic Wireless Power Transfer."*, IEEE International Conference on Computer Communications (**INFOCOM**), 2025.

(25) 2024 Xin-Wei Yao, Wei-Wei Xing, Ke-Chen Zheng, Chu-Feng Qi, Xiang-Yang Li, Qi Song, *" GTDIM: Grid-based Two-stage Dynamic Incentive Mechanism for Mobile Crowd Sensing."*, Pervasive and Mobile Computing (**PMC**), 2024.

(24) 2024 Tian Xie, Qi Song, Qi Zhao, Hao Jiang, Yijie Li, Tongjing Zhu, *"Query Exploration based on Knowledge Reasoning."*, International Conference on Advanced Data Mining and Applications (**ADMA**), 2024.

(23) 2024 Lingchang Kong, Xinyu Wang, Hao Zhou, Fengyu Zhou, Shenyao Jiang, Peide Zhu, Qi Song, Zhi Liu, *"LAORA: Location-Aware Orientation Adjustment for MIMO Magnetic Wireless Charging System"*, IEEE International Conference on Sensing, Communication, and Networking (**SECON**), 2024.

(22) 2023 Miao-Hui Song, Lan Zhang, Mu Yuan, Zichong Li, Qi Song, Yijun Liu and Guidong Zheng, *" CoTel: Ontology-Neural Co-Enhanced Text Labeling"*, ACM The Web Conference (**WWW**), 2023.

(21) 2023 Hanchao Ma, Sheng Guan, Mengying Wang, Qi Song, Yinghui Wu, *"Fair Group Summarization with Graph Patterns"*, IEEE International Conference on Data Engineering (**ICDE**), 2023.

(20) 2023 Zichong Li, Lan Zhang, Mu Yuan, Miao-Hui Song, **Qi Song**, *"Efficient Deep Ensemble Inference via Query Difficulty-dependent Task Scheduling"*, IEEE International Conference on Data Engineering (**ICDE**), 2023.

(19) 2022 **Qi Song,** Mohammad Hossein Namaki, Peng Lin, Yinghui Wu, *"Answering Why-Questions for Subgraph Queries"*, Accepted by IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 34(10), 4636-4649, (2022).

(18) 2021 **Qi Song,** Peng Lin, Hanchao Ma, Yinghui Wu, *"Explaining Missing Data in Graphs: A Constraint-based Approach"*, IEEE International Conference on Data Engineering (**ICDE**), 2021.

(17) 2021 **Qi Song,** Hanchao Ma, Peng Lin, Yinghui Wu, *"GRIP: Constraint-based Explanation of Missing Entities in Graph Search"*, ACM SIGMOD Conference on Management of Data (**SIGMOD(demo)**), 2021.

(16) 2020 Peng Lin**, Qi Song,** Yinghui Wu, *"Repairing Entities using Star Constraints in Multirelational Graphs"*, IEEE International Conference on Data Engineering (**ICDE**), 2020.

(15) 2019 Mohammad Hossein Namaki**, Qi Song,** Yinghui Wu, Shengqi Yang, *"Answering Why-questions by Exemplars in Attributed Graphs"*, ACM SIGMOD Conference on Management of Data (**SIGMOD**), 2019.

(14) 2019 Mohammad Hossein Namaki**, Qi Song,** Yinghui Wu, Jiaxing Pi, *"NAVIGATE: Explainable Visual Graph Exploration by Examples"*, ACM SIGMOD Conference on Management of Data (**SIGMOD (demo)**), 2019.

(13) 2019 **Qi Song,** Mohammad Hossein Namaki, Yinghui Wu, *"Answering Why-Questions for Subgraph Queries in Multi-Attributed Graphs"*, IEEE International Conference on Data Engineering (**ICDE**), 2019.

(12) 2018 **Qi Song,** Bo Zong, Yinghui Wu, Lu-An Tang, Hui Zhang, Guofei Jiang and Haifeng Chen, *"TGNet: Learning to Rank Nodes in Temporal Graphs"*, ACM International Conference on Information and Knowledge Management (**CIKM**), 2018.

(11) 2018 Peng Lin, **Qi Song,** Jialiang Shen, Yinghui Wu, *"Discovering Graph Patterns for Fact Checking in Knowledge Graphs"*, International Conference On Database Systems for Advanced Applications(**DASFAA**), 2018.

(10) 2018 Bo Zong, **Qi Song,** Martin Renqiang Min, Wei Cheng, Cristian Lumezanu, Daeki Cho, Haifeng Chen, *"Deep Autoencoding Gaussian Mixture Model for Unsupervised Anomaly Detection"*, International Conference on Learning Representations(**ICLR**), 2018.

(9) 2018 **Qi Song,** Yinghui Wu, Peng Lin, Xin Luna Dong, Hui Sun, *"Mining Summaries for Knowledge Graph Search"*, IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 30(10), 1887-1900, (2018).

(8) 2018 Peng Lin, **Qi Song,** Yinghui Wu, *"Discovering Patterns for Fact Checking in Knowledge Graphs"*, Journal of Data and Information Quality(**JDIQ**), 2018.

(7) 2018 Peng Lin, **Qi Song,** Yinghui Wu, *"Fact Checking in Knowledge Graphs with Ontological Subgraph Patterns"*, Data Science and Engineering(**DSE**), 2018.

(6) 2018 Wei Cai, Baochun He, Min Hu, Wenyu Zhang, Deqiang Xiao, Hao Yu, Nan Xiang, Jian Yang, **Qi Song,** Songsheng He, Yaohuan Huang, Wenjie Huang, Fucang Jia, Chi-hua Fang, *"A radiomics-based Nomogram for the Preoperative Prediction of Posthepatectomy Liver Failure in Patients with Hepatocellular Carcinoma"*, Surgical Oncology(**SO**), 2018.

(5) 2017 Mohammad Hossein Namaki, Yinghui Wu, **Qi Song,** Peng Lin, Tingjian Ge, *"Discovering Temporal Graph Association Rules"*, ACM International Conference on Information and Knowledge Management (**CIKM**), 2017.

(4) 2017 **Qi Song,** Mohammad Hossein Namaki, Peng Lin, Yinghui Wu, *"Parallel Graph Summarization for Knowledge Search"*, 13th International Workshop on Mining and Learning with Graphs (**MLG**), 2017.

(3) 2016 **Qi Song,** Yinghui Wu, Xin Luna Dong, *"Mining Summaries for Knowledge Graph Search"*, IEEE International Conference on Data Mining (**ICDM**), 2016.

(2) 2014 **Qi Song,** Bo Li, Weiren Yu, Jianxin Li, Bin Shi, *"NSLPA: A node similarity based label propagation algorithm for real-time community detection"*, IEEE International Conference on Utility and Cloud Computing (**UCC**), 2014.

(1) 2013 Jianxin Li, **Qi Song,** Weiren Yu, Chunming Hu, Jian Kang, *"iScreen: A Merged Screen of Local System with Remote Applications in a Mobile Cloud Environment"*, IEEE International Symposium on Service Oriented System Engineering (**SOSE**), 2013.

Patents - before 2020

(6) **Qi Song,** Congmin Min, Jin Guo, Wei Li, *"System and method for intelligent guided shopping"*, US Patent 16,296,169, Status: **Granted**.

(5) Bo Zong, Daeki Cho, Cristian Lumezanu, Haifeng Chen, **Qi Song**, *"Density estimation network for unsupervised anomaly detection"*, US Patent 10,999,247, Status: **Granted**.

(4) Bo Zong, LuAn Tang, **Qi Song,** Biplob Debnath, Hui Zhang, Guofei Jiang, *"Discovering critical alerts through learning over heterogeneous temporal graphs"*, US Patent 10,409,669, Status: **Granted**.

(3) Bo Li, **Qi Song,** Jianxin Li, Weiren Yu, *"Data segmenting method and system of distributed graph calculating system"*, Publication No. : CN104281664B, Status: **Granted**.

(2) Jianxin Li, Weiren Yu, **Qi Song,** Chenggen Sun, Bo Wu, *"Desktop presenting method based on cloud"*, Publication No. : CN103595759B, Status: **Granted**.

(1) Chunming Hu, Weiren Yu, **Qi Song,** Chenggen Sun, Bo Wu, *"File picture sharing method based on cloud"*, Publication No. : CN103595760B, Status: **Granted**.

Awards

2024.06 **2023 ACM China Council Hefei Chapter Rising Star**.

2022.12 **Mozi distinguished youth from USTC**.

2019.04 **NSF Student Travel Award for SIGMOD 2019**.

2019.04 **NSF Student Travel Award for ICDE 2019**.

2017.11 **SIGIR Student Travel Award for CIKM 2017**.

2016.11 **NSF Student Travel Award for ICDM 2016**.

2015 & 2016 **NSF Travel support for NSF Graduate Data Science Workshop 2015, 2016**.

2014.11 **MediaTek Inc. - Beihang Univ. Science and Technology Award**.

Services

SPC	**CIKM'23/25**.
PC	**KDD'25/26,DASFAA'24/25/26,AAAI'23/24/25/26,ECAI'23,etc..**.
Journal Reviewer	**TPAMI, TDSC, TKDE, VLDBJ, JAIR, TNNLS, JDIQ, TBD, JCST, PR, ML, JAIR**.